MATH 110 — Week 1

Notes Chapters 1–2; Homework 1

§1 Sets and functions

The question. When do two descriptions select the same elements? For example, the even elements of $\{1,2,3,4,5,6\}$ are exactly $\{2,4,6\}$.

Definition (Notes Definition 1 and Example 1)**.** A *set* is a collection of distinguishable elements. Write $x \in A$ for membership and $B \subseteq A$ when every element of B belongs to A. The empty set $\varnothing$ has no elements. Relative to an ambient set S, define

$$A^c = \{x \in S : x \notin A\}, \quad A \cup B = \{x : x \in A \text{ or } x \in B\},$$

$$A \cap B = \{x : x \in A \text{ and } x \in B\}, \quad A \setminus B = A \cap B^c.$$

The Cartesian product $A \times B$ consists of ordered pairs (a, b) with $a \in A, b \in B$. Thus $(1,2) \neq (2,1)$.

Definition (Notes Definition 2)**.** A function $f : A \to B$ assigns exactly one value $f(a) \in B$ to every $a \in A$. It is *injective* if $f(a) = f(a')$ implies $a = a'$, *surjective* if every $b \in B$ equals some $f(a)$, and *bijective* if both hold. For $U \subseteq A$ and $V \subseteq B$, its image and inverse image are

$$f(U) = \{f(u) : u \in U\}, \qquad f^{-1}(V) = \{a \in A : f(a) \in V\}.$$

Here $f^{-1}(V)$ is defined even when f is not bijective.

Example (Notes Example 2)

Take $\mathbb{N} = \{1, 2, 3, \ldots\}$. The map $f(n) = n + 1$ is injective: $m + 1 = n + 1$ implies $m = n$. It is not surjective because 1 has no preimage.

Proposition (Notes Chapter 1, finite-set assertion)

If finite sets A, B have the same number n of elements, a function $f : A \to B$ is injective if and only if it is surjective.

Where the proof comes from

Want. Compare distinct inputs with distinct attained outputs.
Know. There are exactly n possible inputs and n possible outputs.
Key step. Count the distinct values among $f(a_1), \ldots, f(a_n)$.

Proof. 1. If f is injective, these are n different elements of the n-element set B, hence exhaust B.
2. If f is not injective, two of the n values coincide, leaving at most $n - 1$ distinct values; it is not surjective.
3. For $n = 0$, the empty function has both properties by their definitions.

□

Testing equality of sets To prove $A = B$, prove $x \in A \Rightarrow x \in B$ and then reverse the roles. For instance, $x \in A \cap (B \cup C)$ means $x \in A$ and either $x \in B$ or $x \in C$. This is equivalent to $x \in (A \cap B) \cup (A \cap C)$; reversing these logical steps proves equality.

Shortcuts

- To disprove injectivity, give two different inputs with the same output; this negates its defining implication.
- To disprove surjectivity, give one codomain element with no preimage; a missing output suffices.
- To compute $f^{-1}(V)$, solve the condition $f(x) \in V$; no inverse function is needed.

Traps

- Equal images need not mean equal inputs: for $f(x) = x^2$ on $\mathbb{R}$, $f(1) = f(-1)$.
- The codomain matters: $x \mapsto x^2$ from $\mathbb{R}$ to $\mathbb{R}$ is not onto, but the same rule with codomain $[0, \infty)$ is onto.

§2 Fields and complex arithmetic

The question. Which scalar arithmetic permits division by every nonzero number? In $\mathbb{Z}$, 2 has no multiplicative inverse; in $\mathbb{Q}$, it has inverse $1/2$.

Definition (Notes Definition 3)**.** A field F has addition and multiplication, distinct elements $0, 1$, and these laws: both operations are associative and commutative; $a + 0 = a$ and $1a = a$; $a(b + c) = ab + ac$; every a has an additive inverse $-a$; every nonzero a has a multiplicative inverse a^{-1}. Subtraction means addition of the additive inverse; division by $a \neq 0$ means multiplication by a^{-1}.

Proposition (Notes Chapter 1, identities, inverses and cancellation)

Identities and inverses are unique. In a field, $a + b = a + c$ implies $b = c$, $a0 = 0$, and $ab = 0$ implies $a = 0$ or $b = 0$.

Where the proof comes from

Want. Justify cancellation from the axioms.
Know. Addition and multiplication are associative and inverses exist in the stated cases.
First try. Dividing by a without checking $a \neq 0$ is not permitted.
Key step. Apply the appropriate inverse; treat $a = 0$ separately.

Proof. 1. Two additive identities satisfy $0 = 0 + 0' = 0'$. Two multiplicative identities satisfy $1 = 1 \cdot 1' = 1'$.
2. If b, c are additive inverses of a, then $b = b + (a + c) = (b + a) + c = c$.
3. If $ab = ac = 1$, then $b = b(ac) = (ba)c = c$. This proves uniqueness of multiplicative inverses.
4. Add $-a$ to $a + b = a + c$ to obtain $b = c$ by associativity and the identity law.
5. Distributivity gives $a0 = a(0 + 0) = a0 + a0$; additive cancellation gives $a0 = 0$.
6. If $ab = 0$ and $a \neq 0$, multiply by a^{-1} to obtain $b = a^{-1}0 = 0$.

□

Example (Notes Example 3)

$\mathbb{Q}$ and $\mathbb{R}$ are fields with their usual arithmetic. In $\mathbb{Q}$, a nonzero p/q has inverse q/p. The integers are not a field: $2n = 1$ has no integer solution.

Definition (Notes Example 4; complex numbers)**.** A complex number has form $a + bi$ with real a, b and $i^2 = -1$. Equality means equality of the real and imaginary parts. Addition is componentwise, and

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

The conjugate of $z = a + bi$ is $\overline{z} = a - bi$.

Proposition (Notes Proposition 3; complex field laws: LADR 1.3)

For complex w, z,

$$\overline{w + z} = \overline{w} + \overline{z}, \quad \overline{wz} = \overline{w}\,\overline{z}, \quad z = \overline{z} \iff z \in \mathbb{R}, \quad z\overline{z} \geq 0.$$

The last quantity is real; it is zero only when $z = 0$. For $z \neq 0$, $z^{-1} = \overline{z}/(z\overline{z})$. In particular $\mathbb{C}$ is a field.

Where the proof comes from

Want. Check conjugation and obtain an inverse.
Know. Multiplication of $a + bi$ uses $i^2 = -1$.
Key step. Multiply $a + bi$ by $a - bi$ to remove its imaginary part.
Example. $(3 + 4i)(3 - 4i) = 25$.

Proof. 1. Write $w = a + bi$, $z = c + di$. Conjugating their sum gives $(a + c) - (b + d)i = \overline{w} + \overline{z}$.

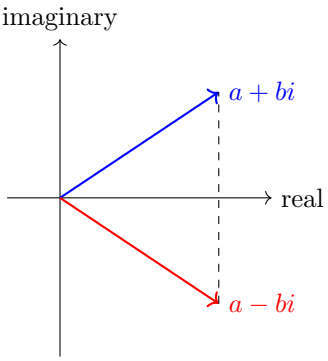
2. Conjugating their product gives $(ac - bd) - (ad + bc)i = (a - bi)(c - di)$.
3. $z = \overline{z}$ is equivalent to $2di = 0$, or $d = 0$.
4. $z\overline{z} = c^2 + d^2 \geq 0$, and it is positive if either real coordinate is nonzero.
5. Thus $z\,[\overline{z}/(c^2 + d^2)] = 1$ for $z \neq 0$. The additive inverse is $-c - di$.
6. Commutativity follows from the displayed multiplication formula; distributivity follows by expanding both coordinates. For associativity, the real and imaginary parts of both $((a+bi)(c+di))(e+fi)$ and $(a+bi)((c+di)(e+fi))$ are respectively $ace - adf - bcf - bde$ and $acf + ade + bce - bdf$.
7. Addition is associative because each real coordinate adds associatively. The real numbers $0, 1$ act as identities. Hence all field axioms hold.

□

Computing powers, products and reciprocals: Notes Example 4 Use $i^2 = -1$ and collect real and imaginary parts:

$$(1 + i)^3 = -2 + 2i, \qquad (5 + i)(4 + 3i) = 17 + 19i, \qquad \frac{1}{3 + 4i} = \frac{3 - 4i}{25}.$$

The notes' cubic example has $(2 + i)^3 = 2 + 11i$ and $(2 - i)^3 = 2 - 11i$. Choosing these particular cube roots gives $(2 + i) + (2 - i) = 4$, and $4^3 - 15 \cdot 4 - 4 = 0$. This computation does not assert a unique complex cube root.



Conjugation changes the sign of the imaginary coordinate and preserves the real coordinate.

Example (Notes Chapter 1, the field $\mathbb{F}_2$)

$\mathbb{F}_2 = \{0, 1\}$ has addition and multiplication modulo 2:

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

.

1. These are integer addition and multiplication followed by taking the remainder modulo 2, so associativity and distributivity descend from the corresponding integer laws.
2. Each element is its own additive inverse; the only nonzero element is 1, with inverse 1. Thus the field axioms hold. Here $1 + 1 = 0$.

A polynomial expression in x is a finite sum of terms $a_k x^k$ with field coefficients, added and multiplied by distributivity and $x^j x^k = x^{j+k}$. The letter x is formal; no numerical value has to be substituted.

Shortcuts

- Compute z^{-1} by conjugating and dividing by the sum of squares of its real coordinates; their product is real and nonzero.
- Reduce powers of i modulo 4, since $i^4 = 1$.
- In $\mathbb{F}_2$, $(x + y)^2 = x^2 + y^2$ because the cross coefficient 2 is zero.

Traps

- $\mathbb{Z}$ is not a field even though addition and multiplication stay in $\mathbb{Z}$: $1/2$ is missing.
- In $\mathbb{F}_2$, cancellation of a nonzero factor works, but division by 2 does not: $2 = 0$ in this field.
- Conjugation preserves products, not their real coordinates separately: $(i)(i) = -1$, although both real parts are zero.

§3 Vector spaces

The question. Which addition and scaling laws are shared by tuples and functions? A sum of two real-valued functions is defined by adding their values at each input.

Definition (Notes Definition 4; notes cite LADR 1.20)**.** A vector space V over a field F is a set with addition $V \times V \to V$ and scalar multiplication $F \times V \to V$. Addition is commutative and associative, has a zero vector 0_V, and every vector has an additive inverse. For $a, b \in F$ and $u, v \in V$,

$$(ab)v = a(bv), \quad a(u+v) = au + av, \quad (a+b)v = av + bv, \quad 1v = v.$$

An element of V is a vector; an element of F is a scalar. We write 0 for either zero when its meaning is clear. Vector subtraction is $u - v = u + (-v)$.

Proposition (Notes Chapter 2; LADR 1.26–1.27 as cited there)
The zero vector and each additive inverse are unique, and vector addition permits cancellation.

Where the proof comes from

Want. Ensure that 0 and $-v$ name single vectors.
Know. Addition has the same identity, inverse and associativity laws used for field addition.
Key step. Repeat the additive arguments, now with vectors.

Proof. 1. For two zero vectors $0, 0'$, $0 = 0 + 0' = 0'$.
2. If $v + u = v + w = 0$, then $u = u + (v + w) = (u + v) + w = w$.
3. If $v + u = v + w$, adding $-v$ to both sides gives $u = w$.

□

Example (Notes Examples 5–7) 1. F^n consists of tuples $(x_1, \ldots, x_n)$ with componentwise addition and scaling; its zero is $(0, \ldots, 0)$. Each axiom holds coordinate by coordinate by a field law. This includes $F = F^1$.
2. For any set S, F^S denotes all functions $S \to F$, with $(f+g)(s) = f(s) + g(s)$ and $(af)(s) = af(s)$. Each axiom holds at each s; the zero function has value 0 everywhere.
3. If $S = \{1, \ldots, n\}$, these functions correspond to tuples; if $S = \mathbb{N}$, they correspond to sequences, written F^∞.
4. $\mathbb{C}$ is a real vector space when only real scalars are allowed. Both $\mathbb{R}$ and $\mathbb{C}$ are rational vector spaces. Their axioms are inherited from the corresponding field laws with the allowed scalars restricted.

Proposition (Notes Proposition 4; notes cite LADR 1.30–1.32)
For $a \in F$ and $v \in V$, $0v = 0_V$, $a0_V = 0_V$, and $(-1)v = -v$.

Where the proof comes from

Want. Derive zero-scaling rules from the axioms, rather than assume them.
Know. Distributivity turns a sum into a sum of vectors; additive cancellation was proved above.
Key step. Write zero as $0 + 0$ and cancel a repeated term.

Proof. 1. $0v = (0+0)v = 0v + 0v$; cancelling $0v$ gives $0v = 0_V$.
2. $a0_V = a(0_V + 0_V) = a0_V + a0_V$; cancelling $a0_V$ gives $a0_V = 0_V$.
3. $(-1)v + v = (-1+1)v = 0v = 0_V$. Uniqueness of the additive inverse gives $(-1)v = -v$.

□

Checking a proposed scalar multiplication On the usual additive set $\mathbb{R}$, define $a * v = 0$ for all a, v. Most scaling identities hold because both sides are zero, but $1 * 1 = 0 \neq 1$, so it is not a vector space. Check the unit axiom explicitly.

Shortcuts

- For a space of tuples, verify an identity in one arbitrary coordinate, which proves it in all coordinates.
- For a function space, verify it at an arbitrary input; functions are equal exactly when all their values agree.
- Check that the operations stay in the proposed set before checking algebraic identities; closure is part of the definition.

Traps

- The scalar field cannot be omitted: $\mathbb{R} \subset \mathbb{C}$ is closed under real scaling but not complex scaling, since $i \cdot 1 = i \notin \mathbb{R}$.
- $0v = 0$ does not alone imply $v = 0$: any nonzero vector multiplied by the zero scalar is a counterexample.

§4 Subspaces, sums and direct sums

The question. When does a condition on vectors survive addition and scaling? The equation $x + y = 0$ does; the equation $x + y = 1$ fails already at the zero vector.

Definition (Notes Definition 5; notes cite LADR 1.34)**.** A subset U of a vector space V is a subspace if $0 \in U$ and $au + v \in U$ whenever $u, v \in U$, $a \in F$. Equivalently it contains zero and is closed under addition and scalar multiplication.

Proposition (Subspace test; Notes Definition 5 and its consequence)

The two formulations of the test agree. A subset satisfying the test is a vector space with the inherited operations.

Where the proof comes from

Want. Avoid rechecking every vector-space axiom for a subset.

Know. The axioms already hold in the ambient space.

Key step. Check closure and the required elements; all remaining identities restrict to the subset.

Proof. 1. The combined condition with $a = 1$ gives addition; with $v = 0$ it gives scalar closure. Conversely the two closure properties give the combined condition.

2. $0 \in U$, and $-u = (-1)u \in U$ by scalar closure and Proposition 4.
3. The inherited associative, commutative, distributive and unit identities hold because they hold for these elements in V.

□

Example (Notes Example 8)

Continuity means that function values approach the value at each point as the input approaches that point; differentiability means that the difference quotient has a limit, denoted $f'(x)$. The closure and derivative rules below are cited from elementary calculus.

1. Within the space of real functions on $(0, 1)$, continuous functions form a subspace: the zero function is continuous and if f, g are continuous and $a \in \mathbb{R}$, then $af + g$ is continuous.
2. Differentiable functions also form a subspace: the zero function is differentiable, and $(af + g)' = af' + g'$.

Example (Notes Example 9)

In $\mathbb{R}^{\mathbb{N}_0}$, where $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$, the sequences satisfying $x_{n+2} = x_{n+1} + x_n$ form a subspace.

1. The zero sequence satisfies the condition. If x, y satisfy it, then

$$(ax + y)_{n+2} = a(x_{n+1} + x_n) + (y_{n+1} + y_n) = (ax + y)_{n+1} + (ax + y)_n.$$

2. The sequence $(0, 1, 1, 2, 3, 5, 8, \ldots)$ is an example. The first two entries determine each next entry by the recurrence, so determine the whole sequence.

Proposition (Notes Proposition 5; notes cite LADR 1C.10)

The intersection $U \cap W$ of two subspaces of V is a subspace.

Where the proof comes from

Want. Keep two sets of conditions simultaneously.

Know. Each condition survives addition and scaling inside its own subspace.

Key step. Apply the subspace test to both sets at once.

Proof. 1. Both subspaces contain 0, hence so does their intersection.
2. If $u, v \in U \cap W$, then $au + v$ lies in U and in W, hence in $U \cap W$.

□

Definition (Sum; Notes Proposition 6)**.** $U + W = \{u + w : u \in U, w \in W\}$. More generally $U_1 + \cdots + U_m$ consists of sums $u_1 + \cdots + u_m$ with $u_j \in U_j$.

Proposition (Notes Proposition 6; notes cite LADR 1.40)
$U + W$ is a subspace and is the smallest subspace containing $U \cup W$.

Where the proof comes from

Want. A subspace containing both U and W.
First try. The union may not contain sums of vectors from the two different sets.
Know. A subspace containing u, w must contain $u + w$.
Key step. Include exactly all such sums and verify closure.

Proof. 1. $0 = 0 + 0$ lies in $U + W$. Also $a(u_1 + w_1) + (u_2 + w_2) = (au_1 + u_2) + (aw_1 + w_2) \in U + W$.
2. Each $u \in U$ equals $u + 0$ and each $w \in W$ equals $0 + w$, so $U \cup W \subseteq U + W$.
3. Any subspace X containing $U \cup W$ contains every $u + w$ by addition closure; hence $U + W \subseteq X$.

□

Definition (Notes Definition 6; two-subspace criterion: LADR 1.46)**.** We abbreviate $u_1 + \cdots + u_m$ by $\sum_{j=1}^{m} u_j$, or $\sum_j u_j$ when the range is clear; $\sum_{j<m}$ means indices $1, \ldots, m-1$. The sum of U, W is *direct*, written $U \oplus W$, when $U \cap W = \{0\}$. A sum of m subspaces is direct when

$$U_k \cap (U_1 + \cdots + U_{k-1}) = \{0\} \qquad (2 \le k \le m).$$

Proposition (Notes Proposition 7; notes cite LADR 1.41, 1.45)
For subspaces $U_1, \ldots, U_m$, the following are equivalent: (i) their sum is direct; (ii) every vector in their sum has a unique representation $u_1 + \cdots + u_m$, $u_j \in U_j$; (iii) $u_1 + \cdots + u_m = 0$ implies $u_1 = \cdots = u_m = 0$.

Where the proof comes from

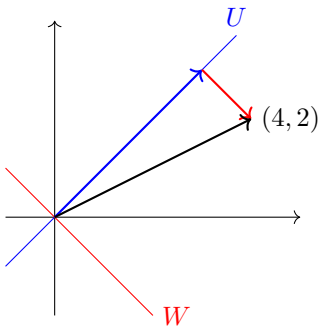
Want. Decide whether a sum determines its summands uniquely.
Know. Subtracting two representations gives a representation of zero.
Key step. Use the last summand's intersection with the preceding sum, then work backwards.
Example. $(x, y) = (x, 0) + (0, y)$ has unique summands from the coordinate axes.

Proof. 1. Under (i), if $\sum_j u_j = \sum_j v_j$, then $u_m - v_m = \sum_{j<m}(v_j - u_j)$ belongs to $U_m \cap \sum_{j<m} U_j = \{0\}$. Hence $u_m = v_m$.
2. Repeat this argument for $m - 1, \ldots, 1$. Existence is the definition of the sum, so (ii) holds.
3. Under (ii), zero already has representation $0 + \cdots + 0$; uniqueness gives (iii).
4. If (i) fails, choose nonzero $u_k = u_1 + \cdots + u_{k-1}$ in the offending intersection. Then $u_1 + \cdots + u_{k-1} - u_k + 0 + \cdots + 0 = 0$ is nontrivial, contradicting (iii).

□

Finding summands without later machinery Let $U = \{(t, t) : t \in \mathbb{R}\}$ and $W = \{(s, -s) : s \in \mathbb{R}\}$.

1. Solving $(x, y) = (t + s, t - s)$ gives $t = (x + y)/2$, $s = (x - y)/2$. Thus every vector is a sum.
2. The only shared vector satisfies $t = s$ and $t = -s$, so $t = s = 0$; the sum is direct.
3. For $(x, y) = (4, 2)$, the summands are $(3, 3)$ and $(1, -1)$.



Adding $(3,3) \in U$ to $(1,-1) \in W$ gives $(4,2)$.

Shortcuts

- To disprove the subspace property, first test zero. The set $x + y = 1$ fails immediately.
- An equation $c_1x_1 + \cdots + c_nx_n = 0$ with fixed field coefficients is preserved by $au + v$: substituting gives $a \cdot 0 + 0 = 0$, checking both closure properties together.
- For two subspaces, test directness by solving $u = w$. For more than two, solve $u_1 + \cdots + u_m = 0$; pairwise intersections do not suffice.

Traps

- A union need not be a subspace: the two coordinate axes contain $(1,0)$ and $(0,1)$ but not their sum $(1,1)$.
- Let $U_1 = \{(t,0)\}$, $U_2 = \{(0,t)\}$, $U_3 = \{(t,t)\}$ (all parameters real). Their pairwise intersections are zero, but $(1,0) + (0,1) + (-1,-1) = 0$, so their sum is not direct.
- For these same lines, $U_1 \cap (U_2 + U_3) = U_1$, while $(U_1 \cap U_2) + (U_1 \cap U_3) = \{0\}$. Intersection does not distribute over subspace sum.

§5 Problem set

HW1 P1. For $f : X \to Y$, $U \subseteq X$, $V \subseteq Y$, prove (a) $f(f^{-1}(V)) \subseteq V$, (b) $U \subseteq f^{-1}(f(U))$. (c) Give strict-inclusion examples and determine exactly when equality holds in each.
Hint: Use the image and inverse-image definitions; distinguish a fixed subset from a statement for every subset.

HW1 P2 (LADR 1A.7). Show that $(-1 + \sqrt{3}\,i)/2$ is a cube root of 1.
Hint: Expand using $i^2 = -1$ and square before multiplying once more.

HW1 P3. For $a, b \in F$, prove $ab = 0$ implies $a = 0$ or $b = 0$.
Hint: Apply the multiplicative-inverse axiom only in the case $a \neq 0$.

HW1 P4 (LADR 1B.2). If $a \in F$, $v \in V$ and $av = 0$, prove $a = 0$ or $v = 0$.
Hint: Use scalar associativity and Notes Proposition 4.

HW1 P5 (LADR 1C.4). For $b \in \mathbb{R}$, let U_b consist of continuous real-valued functions f on $[0,1]$ satisfying $\int_0^1 f(x)\,dx = b$. Prove that U_b is a subspace of $\mathbb{R}^{[0,1]}$ if and only if $b = 0$.
Hint: Test the zero function, then use the subspace test and linearity of the integral (from calculus).

HW1 P6 (LADR 1C.5). Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$?
Hint: The scalar field is $\mathbb{C}$; test multiplication by i.

HW1 P7. In $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$, expand $(x+1)^8$ by first computing $(x+1)^2$ and repeatedly squaring. Explain how this differs from the calculation over $\mathbb{R}$.
Hint: In $\mathbb{F}_2$, the coefficient 2 is zero.

Exam practice (materials/exams/midterm1-review.pdf, Section II, Exercise 7). Prove that the intersection of subspaces is a subspace. What about union?
Hint: Apply Notes Proposition 5, and test the two coordinate axes for union.